ACUTE GASTROENTERITIS IN THE CHILDREN'S EMERGENCY

1. DEFINITIONS

DIARRHEA	↑ in stool frequency (≥ 3 per day) and/or change from baseline stool consistency.
VOMITING	Forceful oral expulsion of gastric contents, associated with contraction of the abdominal, diaphragmatic, and chest wall musculature.
REFLUX	Refers to the passage of gastric contents into the esophagus.
REGURGITATION	Reflux up to the oropharynx or above, without the abdominal & diaphragmatic muscular activity.

ACUTE GASTROENTERITIS (AGE) IS DEFINED AS:

- ↓ or change from baseline consistency of stools **and/or** ↑ in stool frequency (≥ 3 per day)
- ± fever
- ± vomiting

1.1 MICROBIOLOGY

1.1.1 VIRAL

- Vomiting and/or respiratory symptoms are associated with a viral etiology
- Rotavirus is the most severe enteric pathogen, followed by norovirus
- Other enteric viral pathogens include enteric adenoviruses, enteroviruses, astroviruses

	CLINICAL FEATURES
Rotavirus	▫ Associated with more diarrheal episodes, longer lasting diarrhea ▫ Higher clinical severity – severe dehydration & metabolic derangements Benign convulsion with mild gastroenteritis (CwG) "provoked seizures" ▫ Presents on D2 – 3 of AGE (vs. febrile seizures on D1 of illness) ▫ Associated with: ▪ Clustering (≥ 2 episodes) ▪ Presentation with focal seizures / partial seizures ▪ Good prognosis
Norovirus	▫ Increased incidence with rotavirus vaccination ▫ Vomiting is more prominent (compared to other viruses)

1.1.2 BACTERIAL

- Higher fevers (≥ 40 °C), bloody stools, abdominal pain & CNS involvement
- "colitis" pattern – numerous but small volume bloody mucoid stools
- Common pathogens: campylobacter, salmonella
- Most are self-limiting and do not require antimicrobial therapy

2. CLINICAL ASSESSMENT

- Evaluation for clinical features and/or risk factors for etiology (viral vs. bacterial)
- Complications of AGE:
  - Physical – peri-anal excoriations with pain
  - Metabolic – hypoglycemia, electrolyte abnormalities, metabolic acidosis 2' starvation ketosis & HCO_3^- losses
  - Dehydration – see below.

2.1 VOMITING

- Vomiting is a common symptom with a wide differential (CNS, GI, UTI, metabolic causes).
- Evaluate for differentials especially in pre-verbal patients presenting only with vomiting.

2.2 ASSESSMENT OF HYDRATION STATUS

- Clinical signs of dehydration may not be apparent until 3 – 5% dehydration
- Parental reports of normal urine output ↓ the likelihood of dehydration

SYMPTOM	MINIMAL OR NO DEHYDRATION	MILD TO MODERATE DEHYDRATION	SEVERE DEHYDRATION
	(< 3% loss of BW)	(3 – 9% loss of BW)	(≥9% loss of BW)
Mental state	Well; alert	Normal, fatigued, or restless, irritable	Apathetic, lethargic, unconscious
Thirst	Normal	Thirsty; Eager to drink	Drinks poorly
HR	Normal	Normal to ↑	↑
Pulses	Normal	Normal to ↓	Weak, thready, impalpable
Breathing	Normal	Normal; Fast	Deep
Eyes	Normal	Slightly sunken	Deeply sunken
Tears	Present	Decreased	Absent
Mouth / tongue	Moist	Dry	Parched
Skin fold	Instant recoil	Recoil < 2s	Recoil > 2s
CRT	Normal	Prolonged	Prolonged; Minimal
Extremities	Warm	Cool	Cold; Mottled; Cyanotic
Urine output	Normal to ↓	↓	↓↓

3. INVESTIGATIONS

INVESTIGATIONS		REMARKS

Biochemistry	Hypocount
	RP#1/Cl/CO_2
		▫ Consider for infants with the following:
		  ▪ < 2 months old
		  ▪ > 8 diarrheal episodes a day
		  ▪ Family reported signs of severe dehydration (e.g., significant reduced urine output etc.)
		▫ Child with normal HCO_3 is unlikely to be > 5% dehydrated
		▫ KIV capillary blood gas if $HCO_3 \leq 12$ mmol/L

Microbiological	Stool c/s
	Blood c/s
		▫ Routine investigations for etiology not necessary
		▫ Consider viral PCRs for epidemiology or cohorting inpatients (e.g., Norovirus PCR, Rotavirus etc.)
		▫ Consider investigations for:
		  ▪ Patients with underlying chronic conditions/immunocompromised patients
		  ▪ Patients who appear toxic
		  ▪ Patients with suspected bacterial etiology with prolonged symptoms and/or complications in whom treatment is considered

4. MANAGEMENT

4.1 ORAL REHYDRATION FLUID (ORS)

- Fluid of choice for patients with mild – moderate dehydration (> 3% weight loss)
- Preferred over "sports drinks" with high sugar content that contribute to osmotic load which could potentially worsen diarrhea
- 3 available ORS in the Emergency Department:

	Vol.	Na (mmol/L)	K (mmol/L)	Cl (mmol/L)	Glu (mmol/L)	Osm (mOsm/L)
Hydralyte®	62.5ml	55	20	45	80	245
Repalyte®	240ml	20	10	15	165	
#ORS	240ml	75	20	65	75	245

#On eIMR = "ORS (Sodium 75 mmol/L)"

- Hydralyte®
  - Composition follows WHO recommendations
  - Sweetened with natural 0-cal sweetener that does not add to overall osmolality
- Repalyte®
  - More palatable as glucose concentrations are high
  - Composition of other solutes do not follow WHO recommendation
  - Ideal for older pediatric patients (8 – 10 years):
    - Require larger ORS volumes
    - Palatability is important
    - Not mature enough to overlook the taste of standard ORS.

- Rehydration strategy:

DEGREE OF DEHYDRATION	REHYDRATION	REPLACEMENT
Mild – Moderate	50 – 100 ml/kg of ORS over 3 – 4H	With each episode of watery diarrhea or vomiting, serve 2 – 4 sachets of Hydralyte®
Minimal – None	Not required.	

4.2 IV FLUIDS

- Indicated for: shock, severe dehydration with altered conscious level & metabolic acidosis, persistent vomiting despite appropriate fluid administration, severe abdominal distension, and ileus
- Rapid rehydration: IV normal saline or Lactated Ringers **20 ml/kg** over 2 – 4 hours, followed by:
  - Oral rehydration, OR
  - IV infusions

4.3 ANTIBIOTICS

- Should not be given to otherwise healthy children with acute bacterial gastroenteritis
- Indication for antibiotics:

HIGH RISK PATIENTS	▫ Neonates / young infants (< 3 months) ▫ Underlying immunodeficiency (primary or secondary e.g., HIV) ▫ Corticosteroids/immunosuppressive therapy/anatomical or functional asplenia ▫ Inflammatory Bowel Disease (IBD)
SEVERE/COMPLICATED DISEASE	▫ Significant dehydration ▫ Protracted disease ▫ Toxic/unwell looking
SPECIFIC PATHOGENS	▫ Culture proven *Shigella* ▫ Dysenteric *Campylobacter* GE (most efficacious if started within first 3 days of disease) ▫ Enterotoxigenic *E. coli* ▫ *Vibrio cholera*

- Empiric antibiotics:
  - PO azithromycin 10 mg/kg (500 mg) OD x 3 days **OR**
  - PO ciprofloxacin 10 – 20 mg/kg (500 mg) BD x 3 – 5 days

5. ADMISSION CRITERIA

- Moderate/severe dehydration
- Intractable vomiting despite intervention with antiemetics (persistence of spontaneous vomiting without attempting oral feeding → contribute to dehydration with upper GI losses)
- High risk patients
- Diagnostic uncertainty (e.g., surgical abdomen)

6. DISCHARGE ADVICE FOR PARENTS

6.1 HYDRATION ADVICE

- Frequent small volume feeds via spoon/syringe (e.g., 5 ml every 5 min with gradual increments)
    - Child would be very thirsty and tend to drink very quickly
- Aim to restart normal diet after rehydration therapy
- Goal is to prevent dehydration – if patient can compensate / replace diarrheal losses with no vomiting, patient can be managed at home.
- Features that child is adequately hydrated:
    - Activity level
    - At least 4 wet diapers a day

6.2 PERIANAL EXCORIATION

- 2 reasons for perianal excoriation:
    - Contact dermatitis from infrequent diaper change
    - Abrasions from by wiping resulting in significant pain from exposed nerve endings
- Advice for parents on perineal hygiene:
    - Splash water instead of wiping skin with wet wipes – dab dry with a clean towel
    - Apply lignocaine gel for pain relief (if desquamation of skin has already occurred)
    - Apply a thick layer of Zinc oxide cream (Desitin® cream), the cream works by:
        - Forming a protective barrier on the skin to reduce friction
        - Blocking wetness from the skin
        - Protecting against enzymes found in feces
        - Protects and soothes irritated skin
    - Ensure diapers are appropriately sized and not applied too tightly, this may result in friction

ANNEX A: OVERVIEW OF MEDICATIONS

DRUG	ROUTE	DOSE	REMARKS
ANTI-EMETIC			
Ondansetron	SL	**0.15 mg/kg** ONCE	▫ Benefits: ↓ risk of persistent vomiting, ↓ need for IV fluids, ↓ need for hospital admission ▫ Side effects: ▪ Worsen diarrhea ▪ Risk of malignant arrhythmias in patients with hypokalemia and hypomagnesemia or long QT syndrome
ANTI-DIARRHEAL			
Smecta® (Diosmectite)	PO	½ sachet BD (≤ 2-years) 1 sachet BD (> 2-years)	▫ ↓ stool frequency and duration of diarrhea. ▫ Do not discharge with other medications.
Hidrasec® (Racecadotril)	PO	**1.5 mg/kg** TDS • Round to the nearest sachet size (10mg/30mg) • Max. 60mg • Not more than 7 days	▫ Mechanism ▪ Anti-enkephalinase, resulting in ↑ enkephalins ▪ Inhibits secretion in the gut with little effect on motility ▫ Mitigates rotavirus-induced & cholera toxin-induced secretory diarrhea (in animal models)
PROBIOTICS			
LactoGG® (*L rhamnosus* GG)	PO	½ capsule OD (< 1-year) 1 capsule OD (≥ 1-year)	▫ Probiotics as a group, reduces duration of diarrhea by approximately 1 day. ▫ Effects are strain specific ▫ ESPGHAN recommendation ▪ Strong: *L rhamnosus* GG & *S boulardii* ▪ Weak: *L reuteri* DSM 17938 & *L acidophilus* LB
Lacteol forte® (*L acidophilus* LB)	PO	½ sachet BD (≤ 2-year) 1 sachet BD (> 2-year)	
NOT RECOMMENDED IN PEDIATRIC PATIENTS			
Anti-motility	▫ Loperamide is not recommended ▫ Contraindicated in: Patients < 2-years, bacterial GE, acute dysentery ▫ Adverse effects: ▪ Risk of cardiac arrest, torsades de pointes and death with higher than recommended doses ▪ Prolongs gut transit time → ↑ bacterial growth, ↑ diarrheal losses from osmotic load ▫ Paralytic ileus → abdominal pain & distension		

7. REFERENCES

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